

CAD CAM Database and Data exchange

Automation

Notes on Line Balancing and LCR

CAD CAM Database

CAD CAM Database

- Computer-aided design can increase the productivity of the design department in the company
- CAD system offers the opportunity to develop the data base needed to manufacture the product
- In the conventional manufacturing cycle - *engineering drawings were prepared by design draftsmen* and then used by *manufacturing engineers to develop the process plan*
- The activities involved in designing the product were separated from the activities associated with process planning.
- The two-step procedure was employed was time consuming and involved duplication of effort by design and manufacturing persons.
- In an *integrated CAD/CAM system, a direct link is established between product design and manufacturing.*

CAD CAM Database

It includes all the data on the product generated during design (geometry data, bill of materials and parts lists, material specifications, etc.) integrated with data required for manufacturing

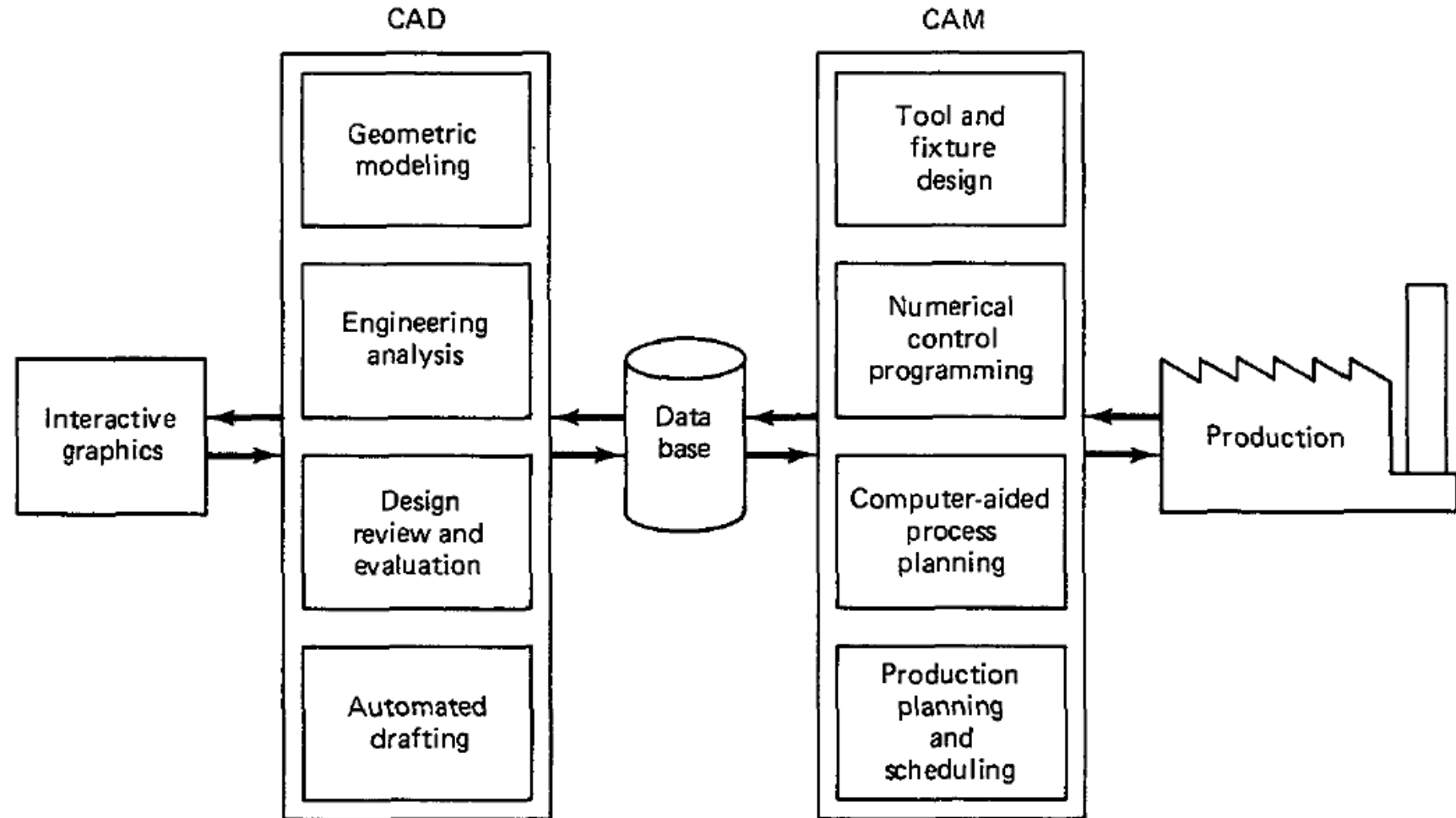


FIGURE 4.10 Desirable relationship of CAD/CAM data base to CAD and CAM.

Benefits of CAD

1. Improved engineering productivity
2. Shorter lead times
3. Reduced engineering personnel requirements
4. Customer modifications are easier to make
5. Faster response to requests for quotations
6. Avoidance of subcontracting to meet schedules
7. Minimized transcription errors
8. Improved accuracy of design
9. In analysis, easier recognition of component interactions
10. Provides better functional analysis by prototype testing
11. Assistance in preparation of documentation
12. Designs have more standardization
13. Better designs provided
14. Improved productivity in tool design
15. Better knowledge of costs provided
16. Reduced training time for routine drafting tasks and NC part programming
17. Fewer errors in NC part programming
18. Provides the potential for using more existing parts and tooling
19. Helps ensure designs are appropriate to existing manufacturing techniques
20. Saves materials and machining time by optimization algorithms
21. Provides operational results on the status of work in progress
22. Makes the management of design personnel on projects more effective
23. Assistance in inspection of complicated parts
24. Better communication interfaces and greater understanding among engineers, designers, drafters, management, and different project groups

CAD Data Exchange

- Use CAD model data for various CAD based engineering applications like production drawings, 3Dprinting, FEA, CFD, CAE, CAM etc.
- Geometric data exchange refers to conversion from one geometric data format to another.
- CAD/CAM software developers use different formats to store the data.
- *Fundamental incompatibilities among entity representations further complicate exchanging the modelling data.*
- The Geometry data format in which part geometry information is stored often varies from one CAD/CAM system to another and hence the *format need not be unique one*

Need for geometric data exchange

Heterogenous expertise in industry

- In one organization expertise is available in one CAD package while in other organization people are more familiar with other package.
- This heterogeneity will always exist as long as number of alternative packages are available in market.
- That arises the need of data exchange.

Use of application specific packages

- Single package may not satisfy all the requirements of design and manufacturing.
- Even if only one versatile comprehensive package is used for all CAD/CAM operations, still data exchange is needed from one module to another.

Need for geometric data exchange contd.

- Migration from one system to another
 - May be because of availability of better package in the market.
 - Lack of technical support from the parent company.
- Data exchange with collaborators/ customers/suppliers
 - Because of technology transfer
 - Data exchange between developer and beneficiary
- Rapid pace of technological change

Data that needs to be exchanged

Shape data

- Geometrical Information
- Topological Information
- Attributes such as font, colour, material
- Annotations

Non-shape data

- It includes graphics data such as shaded images
- Measuring units
- Resolution

Data that needs to be exchanged contd.

Design data

- Information that designers generate from geometric models for analysis purpose
- Mass property and finite element mesh data belong to this type of data.

Manufacturing data

- It consists of information as tooling, NC tool path,
- Tolerancing, process planning, tool design and bill of materials

Options available for data exchange

- Native format
- Neutral format
- Standard format
- Binary format
- ASCII format

Options available for data exchange

Native Format:

- This is Company's own format
- Generally not disclosed
- Always with some specific extension.
- File can be read only by the same package.
- It is not a standard.
- It is not opened by another package unless there is a translator.

Options available for data exchange

Neutral format:

- package gives an optional format which can be very easily interpreted by the outside world.
- Generally it comes with all information like what type of data is stored , what is the order of information etc...
- It may come with manual wherein everything is mentioned about how data is stored.

Standard format:

- Accepted by national or international standard.

Options available for data exchange

Binary format

A binary file is a computer file that is not a text file.

ASCII format

- American *Standard Code for Information Interchange*
- It is a standard encoding for alphanumeric characters used in computers and related devices.
- ASCII was introduced by the United States of America Standards Institute (USASI) now known as the American National Standards Institute.

ASCII - Binary Character Table

Letter	ASCII Code	Binary
A	065	01000001
B	066	01000010
C	067	01000011
D	068	01000100

Different formats

Some standard available formats are-

- IGES, DXF, STL, STEP, PDES, ACIS, Parasolid
- DXF- Drawing Exchange Format
- IGES- Initial Graphics Exchange Format
- STL- Standard Triangular Language
- STEP- Standard for Exchange of Products data

IGES (Initial Graphics Exchange Specification) format

- It is the first standard exchange format
- IGES has been revised a few times since its version 1.0 was released in 1980
- The various IGES versions share common characteristics
- Each version must remain upwards compatible with previous version
- This means that a processor that is fully conforming to the latest version can correctly interpret IGES files written in accordance with prior versions

IGES (Initial Graphics Exchange Specification) format

- CAD/CAM vendors or companies specialized in database transfer must write software to translate from their systems to IGES format and vice versa.
- The software that translates from the native database format of a given CAD/CAM system to the IGES format is called a *preprocessor*
- The software that translates in a opposite way (from IGES to a CAD CAM system) is called as *postprocessor*
- The preprocessors and postprocessors are also called as translators

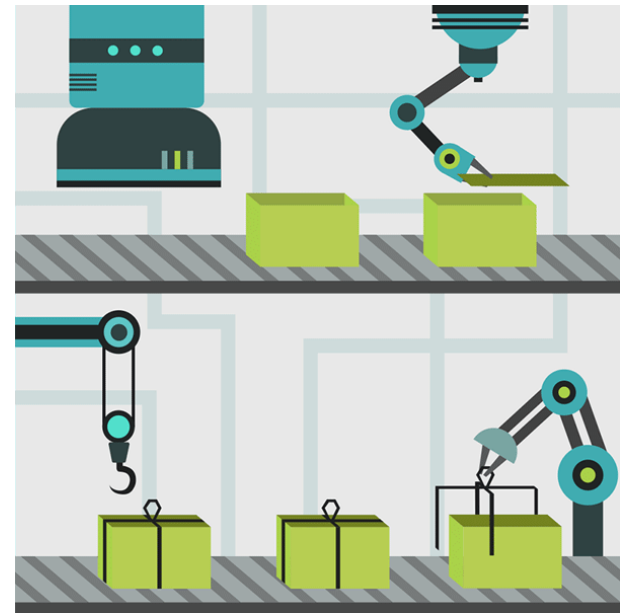
Automation



- Automation is the use of technology to perform tasks with reduced human assistance
- Automation is the technology by which a process or procedure is accomplished without human assistance



- Automation can be applied in a wide variety of areas, it is most closely associated with the manufacturing industries
- Highly automated systems, there is virtually no human participation
 - automated machine tools that process parts
 - transfer lines that perform a series of machining operations
 - automated assembly systems
 - manufacturing systems that use industrial robots to perform processing or assembly operations
 - automatic material handling and storage systems to integrate manufacturing operations
 - automatic inspection systems for quality control



ELEMENTS OF AUTOMATED SYSTEM

- Automated system implemented using a *program of instructions* combined with a *control system* that executes the instructions.
- To automate a process, *power* is required, both to drive the process itself and to operate the program and control system

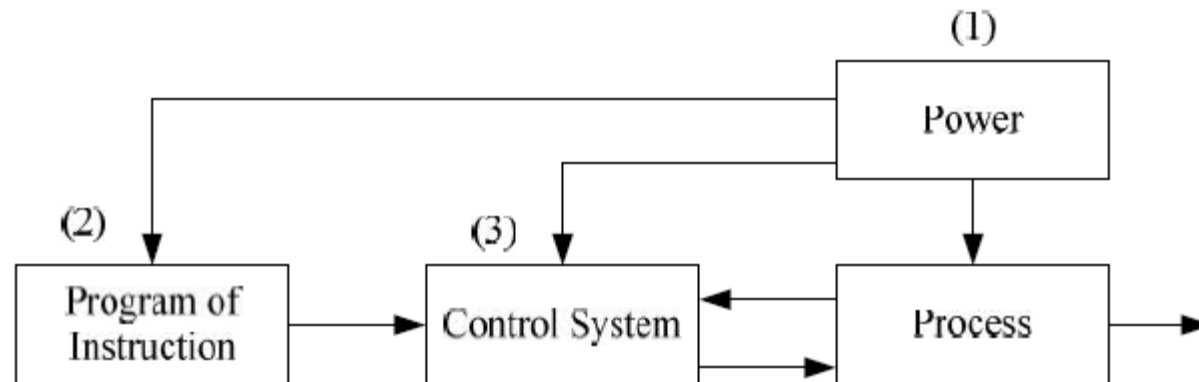


Figure 1.1 Elements of an automated system

Power

An automated system is used to operate some process, and power is required to drive the process as well as the controls. The principal source of power in automated systems is electricity.

Program of Instructions

The actions performed in an automated process are defined by a program of instructions.

POI consist of :

- **Work Cycle Programs:** *The processing steps for the work cycle* are specified in a work cycle program. work cycle consists of essentially *one step, which is to maintain a single process parameter*. However, the system becomes *complicated consisting of multiple steps with more number of process parameters* are required to control the system
- **Process parameters:** *are inputs to the process* (process variables) such as temperature setting of a furnace etc.

Control System

The control system of the automated system executes the program of instructions. *The control system causes the process to accomplish its defined function which is to carry out some manufacturing operation.*

The controls in an automated system can be either ***closed loop (feedback control system)*** and ***open loop***

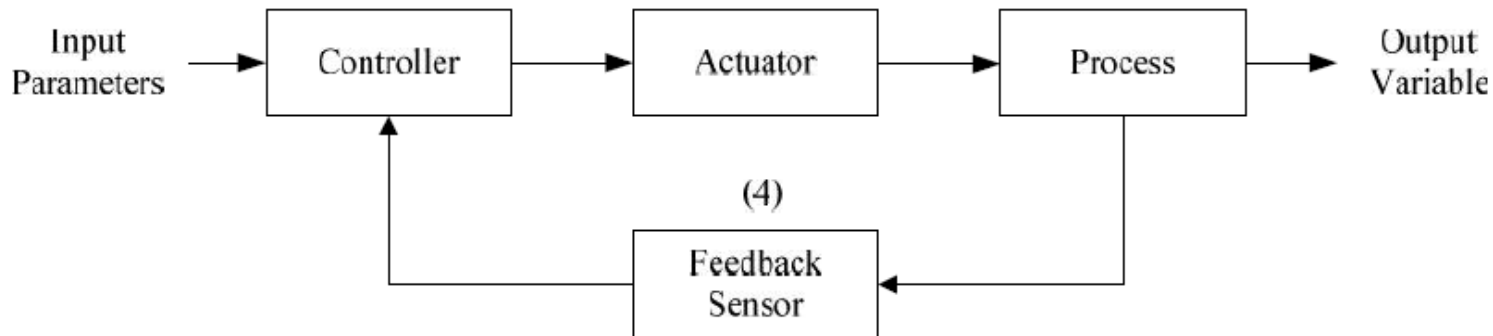


Figure 1.2 A feedback control system

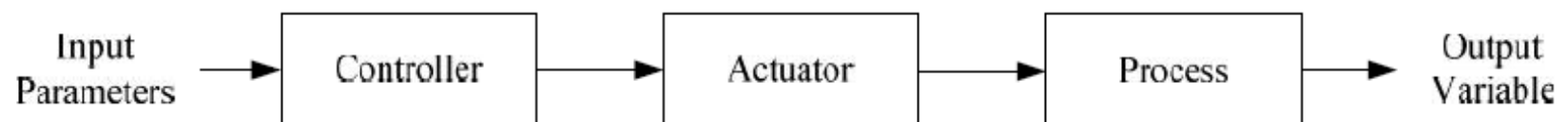


Figure 1.3 An open loop control system

Levels of automation

Levels of automation can be classified based on the complexity of the system being automated. Here's a breakdown of the different levels

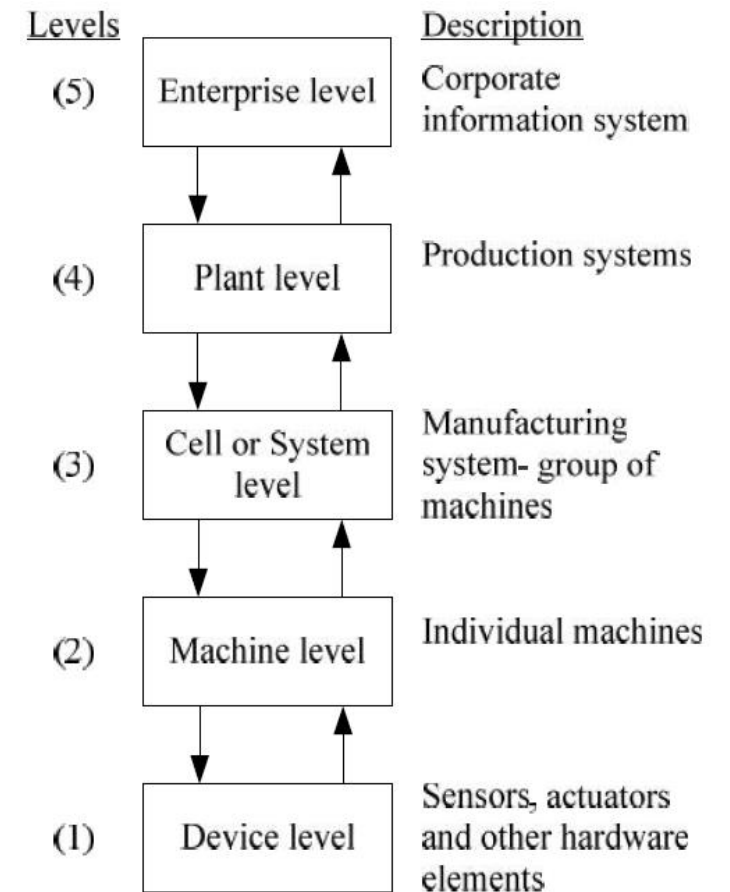


Figure 1.4 Level of automation

1. Device Level Automation

- This is the most basic level of automation, where individual devices are automated to perform specific tasks.
- Examples include automatic thermostats, light dimmers, and self-cleaning ovens.
- In this level, sensors collect data on the environment (e.g., room temperature) and send signals to controllers (e.g., thermostat) to adjust settings (e.g., adjust furnace output) to maintain a desired state (e.g., comfortable room temperature).

2. Machine Level Automation

- *This level involves automating a single machine to perform a sequence of operations.*
- Programmable logic controllers (PLCs) are typically used to control these machines.
- Examples include industrial robots in assembly lines, CNC machines, and automated packaging machines.
- Here, the automation is more complex than device level. Machines can perform a series of steps without human intervention. Sensors provide feedback on the machine's operation

3. Cell or System Level Automation

- This level automates a group of interconnected machines or devices working together to achieve a common goal
- It involves coordinating the operation of multiple machines and may involve material handling systems and automated transfer mechanisms.
- Examples include robotic welding cells, automated painting systems, and automated assembly lines.
- This level integrates multiple machines into a system that can perform a complex task. Communication and coordination between machines are crucial for efficient operation.

4. Plant Level Automation

- This level involves automating the entire production process within a factory or plant.
- It integrates all the automation systems at the cell or system level, along with production planning, scheduling, and inventory management systems.
- Plant level automation provides a holistic view and control over the entire production process. It optimizes production efficiency, resource allocation, and ensures smooth operation throughout the plant

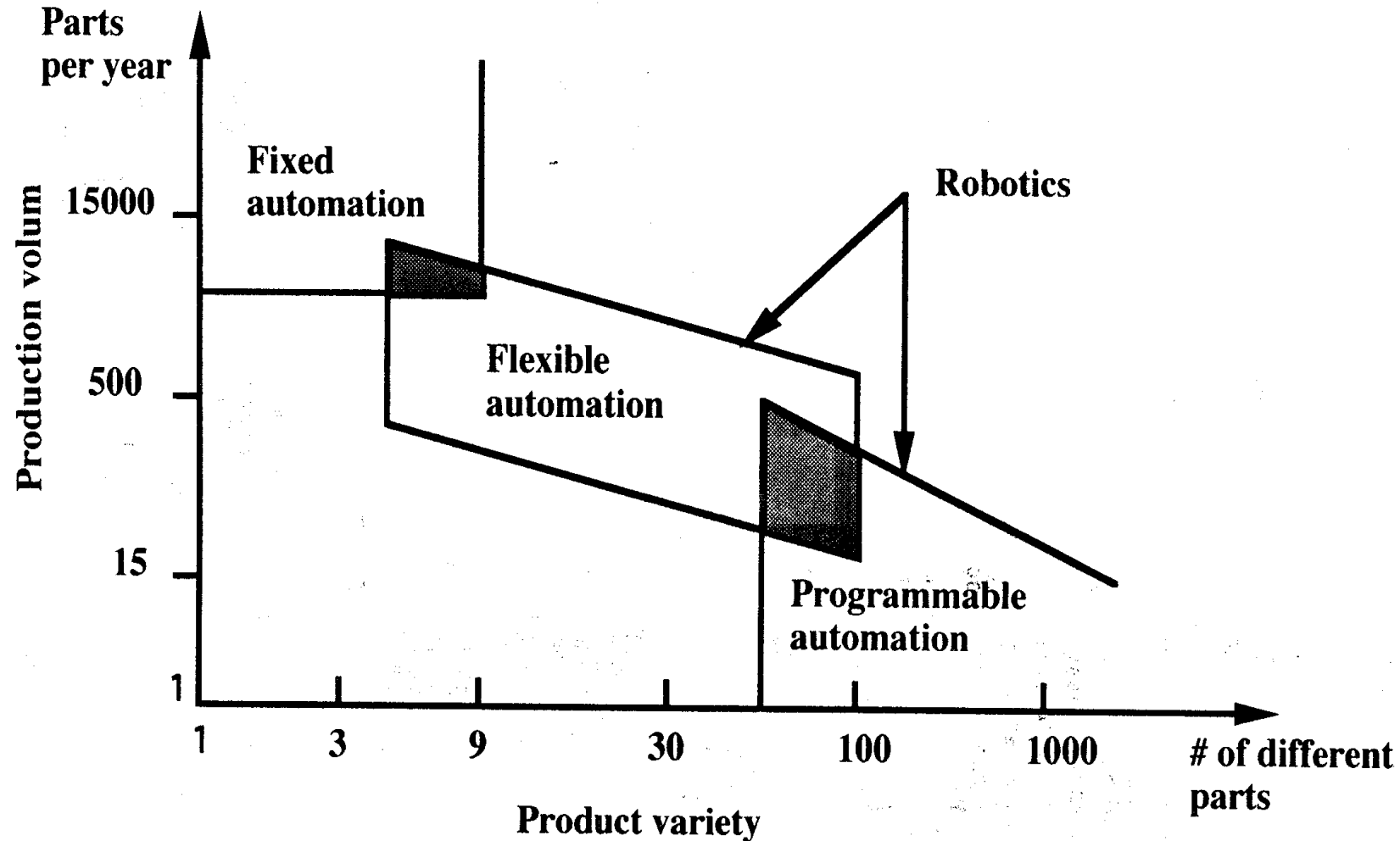
5. Enterprise Level Automation

- This is the highest level of automation, where all the automation systems across an entire enterprise are integrated.
- It includes not only the production process but also other enterprise functions such as supply chain management, customer relationship management (CRM), and enterprise resource planning (ERP).
- Enterprise level automation provides an integrated view and control over all aspects of the business.

Three Basic Types of Automation

- **Fixed automation** - the processing or assembly steps and their sequence are fixed by the equipment configuration
- **Programmable automation** - equipment is designed with the capability to change the program of instructions to allow production of different parts or products
- **Flexible automation** - an *extension of programmable automation* in which there is virtually no lost production time for setup changes or reprogramming

Three Basic Types of Automation



Features of Fixed Automation

- High initial investment for specialized equipment
- High production rates
- *The program of instructions cannot be easily changed because it is fixed by the equipment configuration*
 - Thus, little or no flexibility to accommodate product variety





Features of Programmable Automation

- High investment in general purpose equipment that can be **reprogrammed**
- *Ability to cope with product variety by reprogramming the equipment*
- Suited to batch production of different product and part styles
 - Lost production time to reprogram and change the physical setup
- *Lower production rates than fixed automation*

Features of Flexible Automation

- High investment cost for custom-engineered equipment
- *Capable of producing a mixture of different parts or products without lost production time for changeovers and reprogramming*
 - Thus, continuous production of different part or product styles
- *Medium production rates*



AUTOMATED FLOW LINES

- An automated flow line consists of several machines or workstations which are linked together by work handling devices that transfer parts between the stations
- A raw work-part enters *one end of the line* and the processing steps are performed sequentially as the part moves from one station to the next
- It is possible to incorporate buffer storage zones into the flow line, either at a single location or between every workstation.
- It is also possible to include inspection stations in the line to automatically perform intermediate checks on the quality of the work-parts

objectives of the use of flow line automation

- To reduce labor costs
- To increase production rates
- To reduce work-in-process
- To minimize distances moved between operations
- To achieve specialization of operations
- To achieve integration of operations

Configurations of automated flow line - In-line type

- The in-line configuration consists of a sequence of workstations in a more or less straight line arrangement*

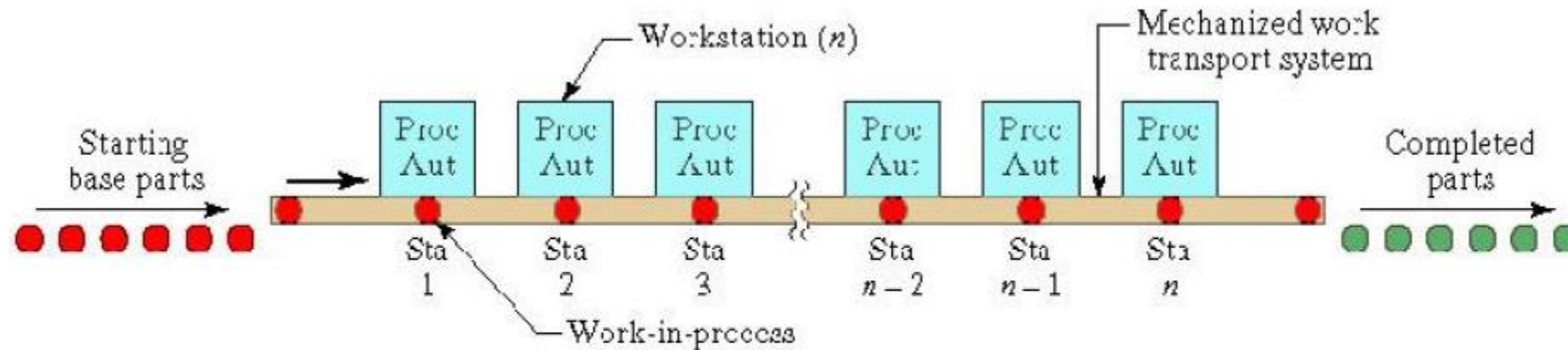


Figure 1.8 Flow line (Inline type)

Segmented In-Line Type

- The segmented in-line configuration consists of two or more straight line arrangement which are usually perpendicular to each other with L shaped or U shaped or rectangular shaped
- *The flow of work can take a few 90° turns, either for workpiece reorientation, factory layout limitations*

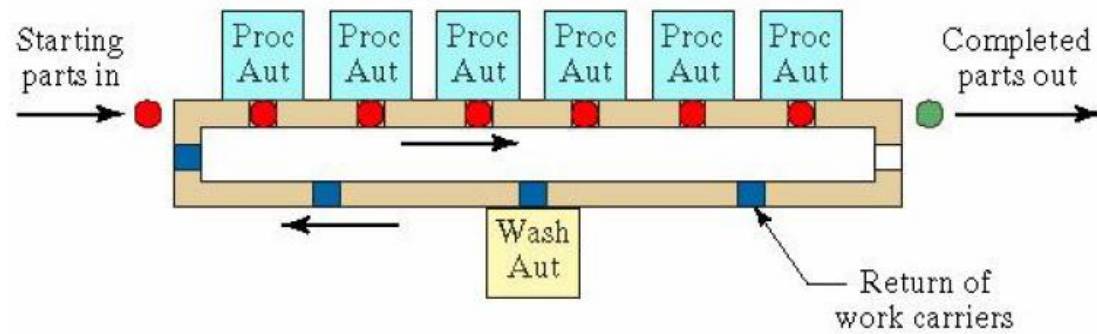


Figure 1.14 Rectangular-shaped configuration

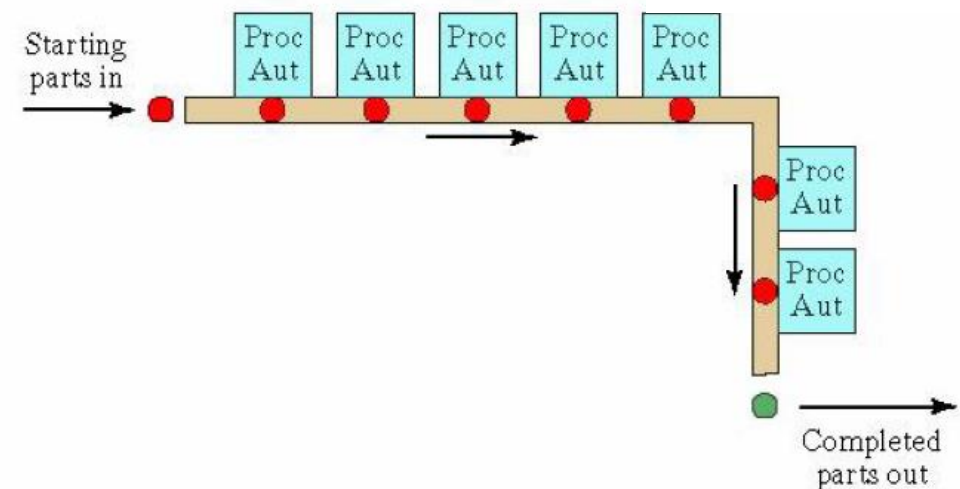


Figure 1.12 L-shaped configuration

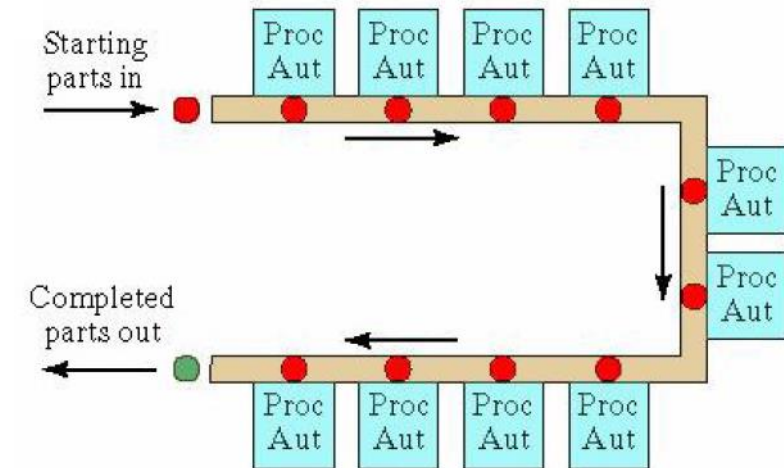


Figure 1.13 U-shaped configuration

Rotary type

- The work-parts are indexed around a circular table or dial
- The workstations are stationary and usually located around the outside periphery of the dial.
- The parts ride on the rotating table and positioned, in turn, at each station for its processing or assembly operation.

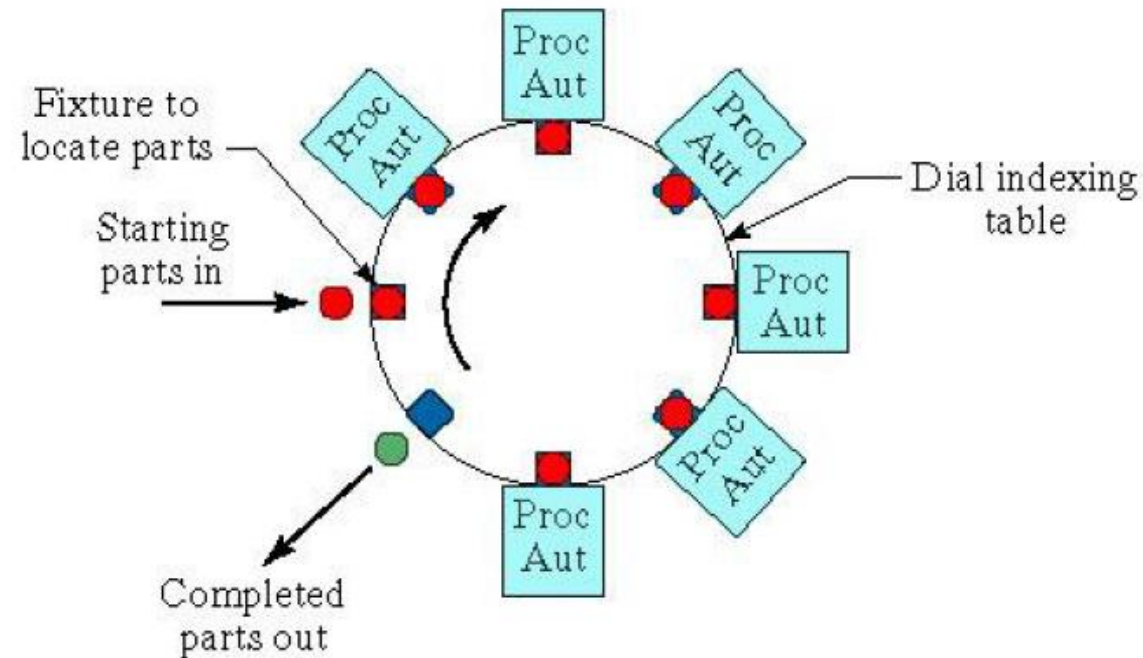


Figure 1.15 Rotary configuration

Transfer mechanisms

- To move parts between stations
- These mechanisms can be grouped into two types:

Linear travel for in-line machines

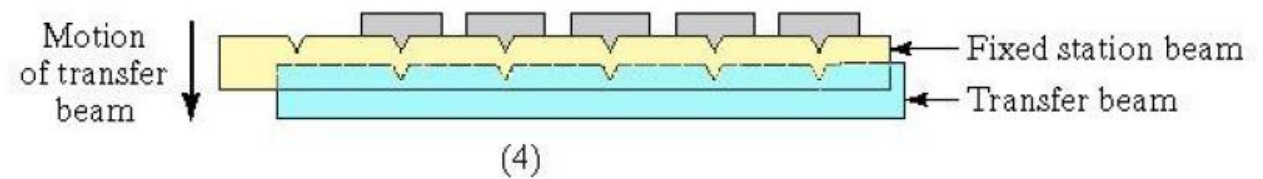
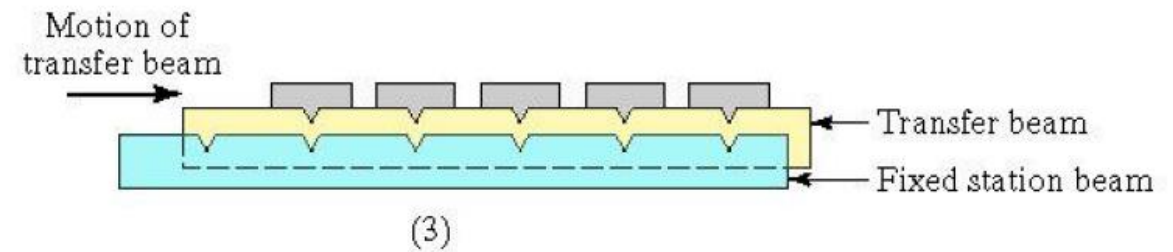
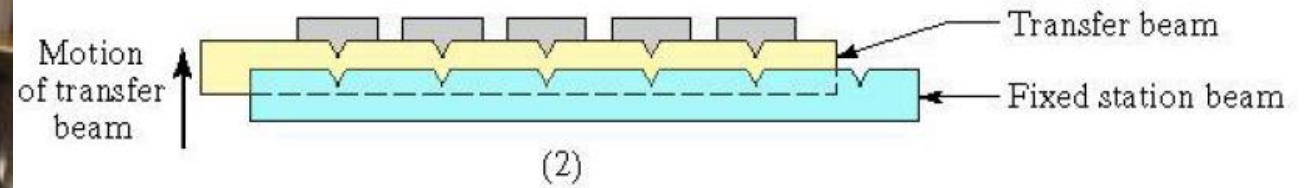
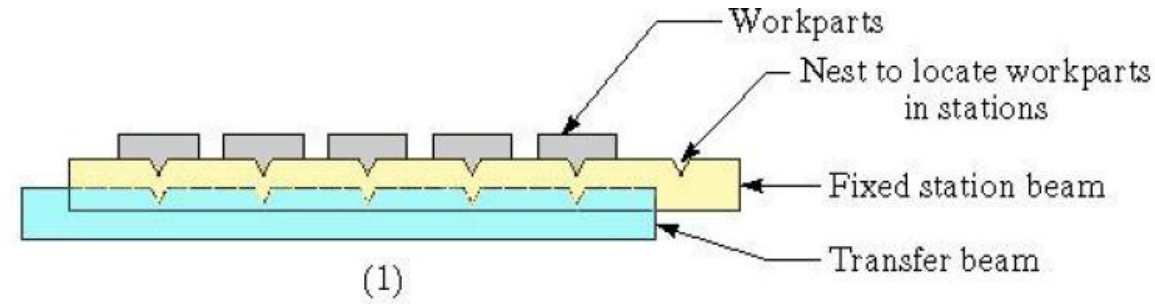
Rotary motion for dial indexing machines

Linear transfer mechanisms

a. Walking beam systems

- The work-parts are lifted up from their workstation locations by a transfer bar and moved one position ahead, to the next station.
- The transfer bar then lowers the parts into nests which position them more accurately for processing.
- The motion of the beam is generated by a rotating camshaft powered by an electric motor

a. Walking beam systems



b. Powered roller conveyor system

- The conveyor can be used to move pans or pallets possessing flat riding surfaces
- The rollers can be powered by either of two mechanisms.

Belt drive

Chain drive

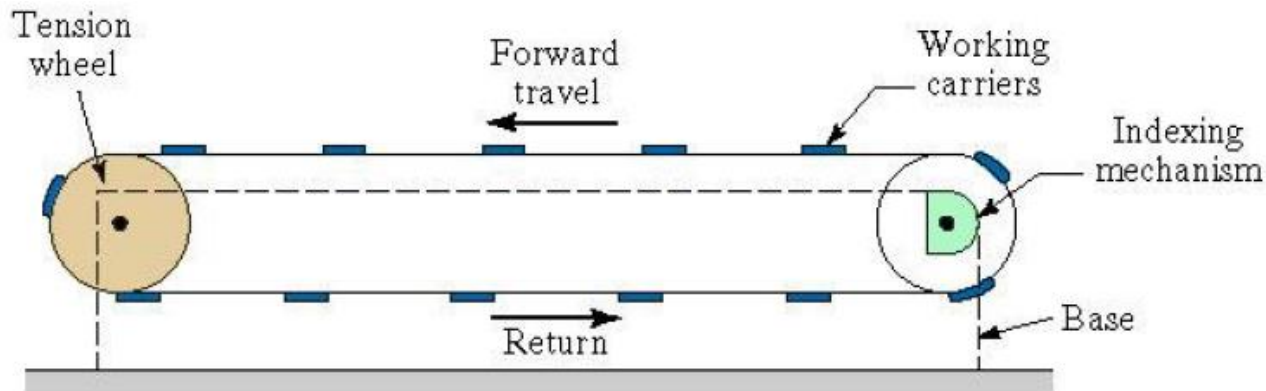


Figure 1.19 Chain drive conveyor

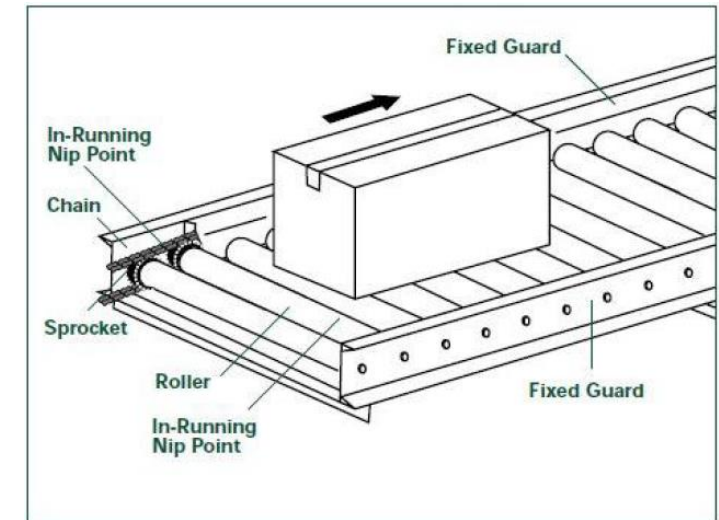


Figure 1.18 Powered roller conveyor

Rotary transfer mechanisms

a. Rack and pinion

- This mechanism is simple but is not suitable to the high-speed operation
- Piston to drive the rack, which causes the pinion gear and attached indexing table to rotate

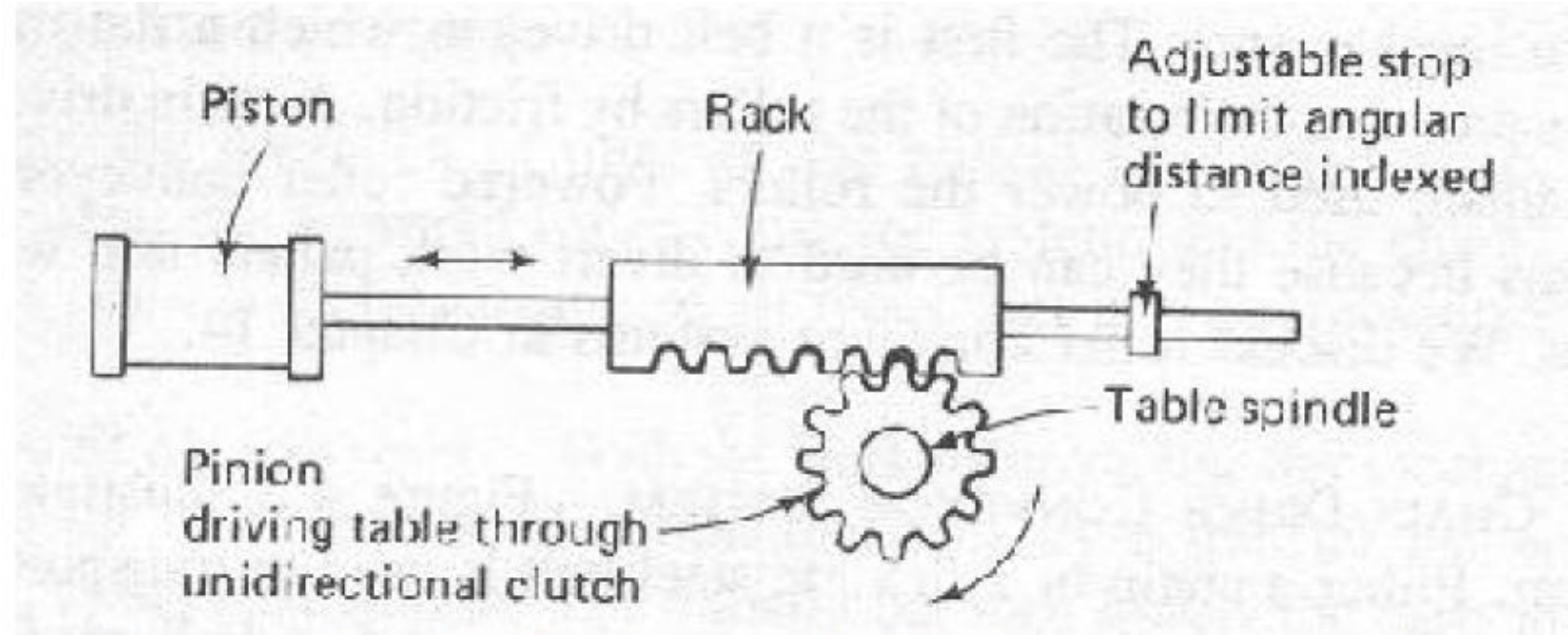
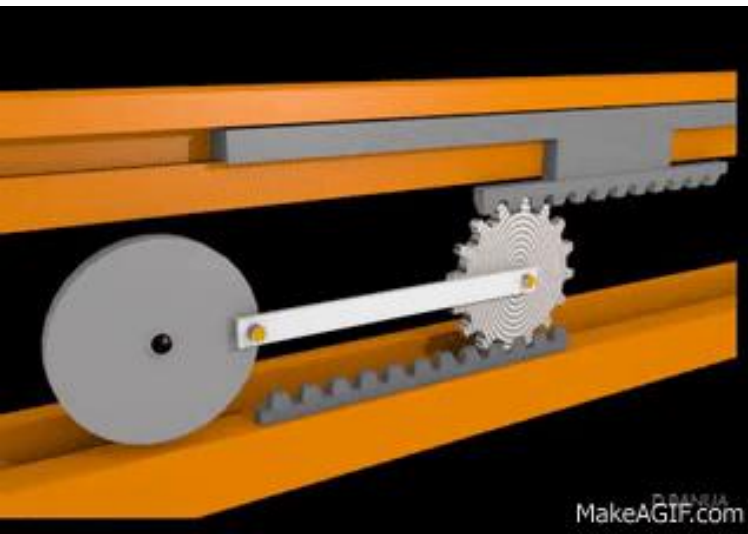
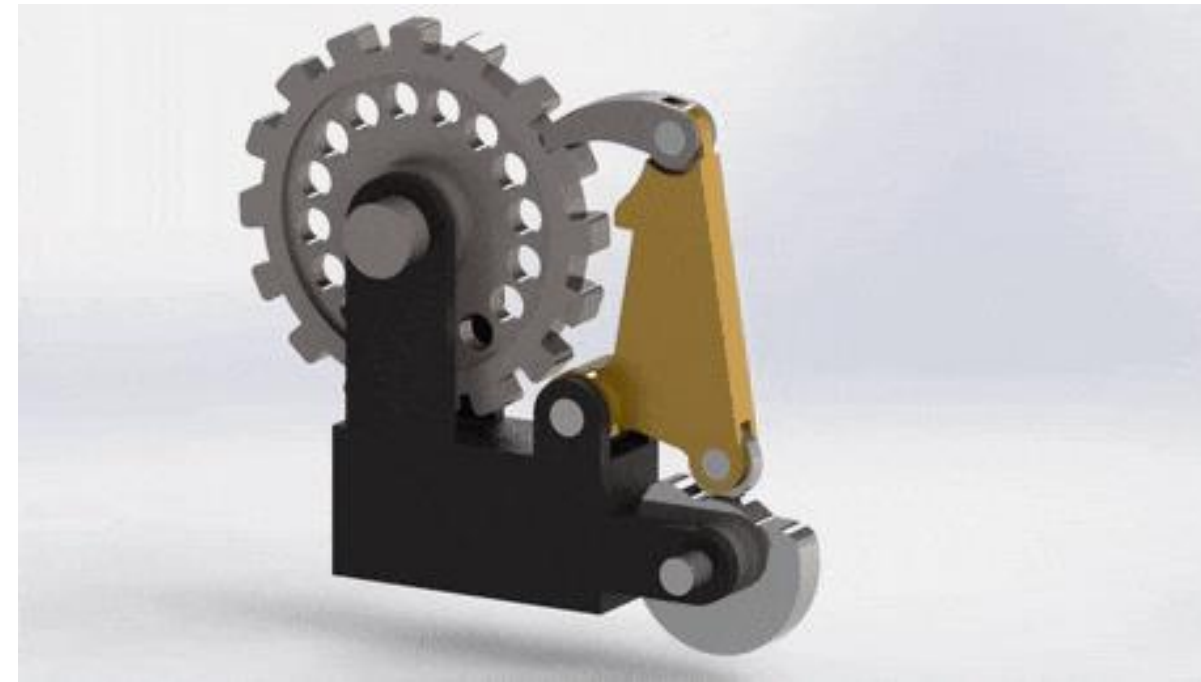
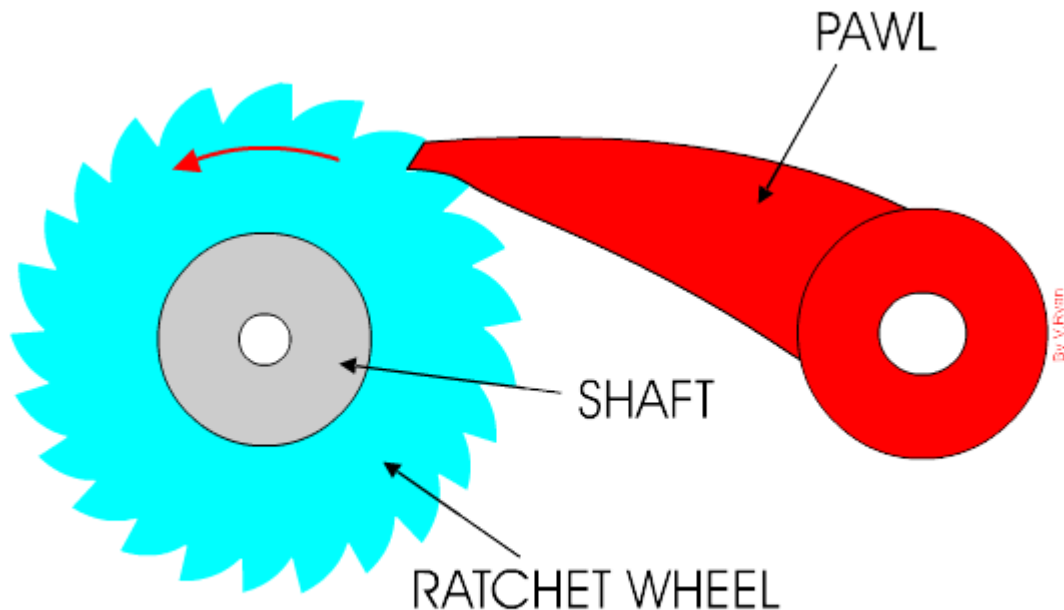
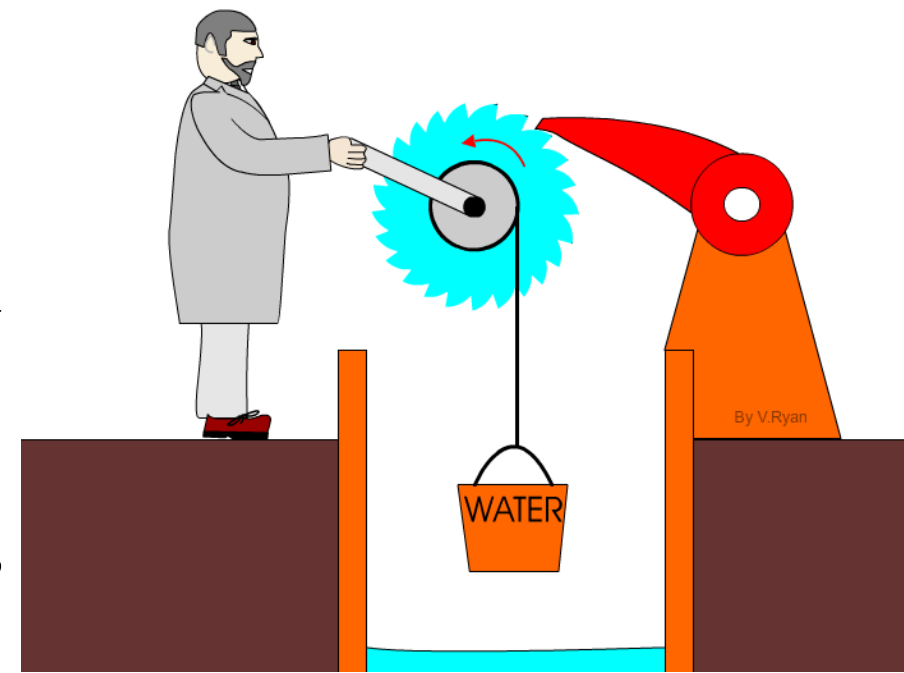


Figure 1.20 Rack and pinion mechanism

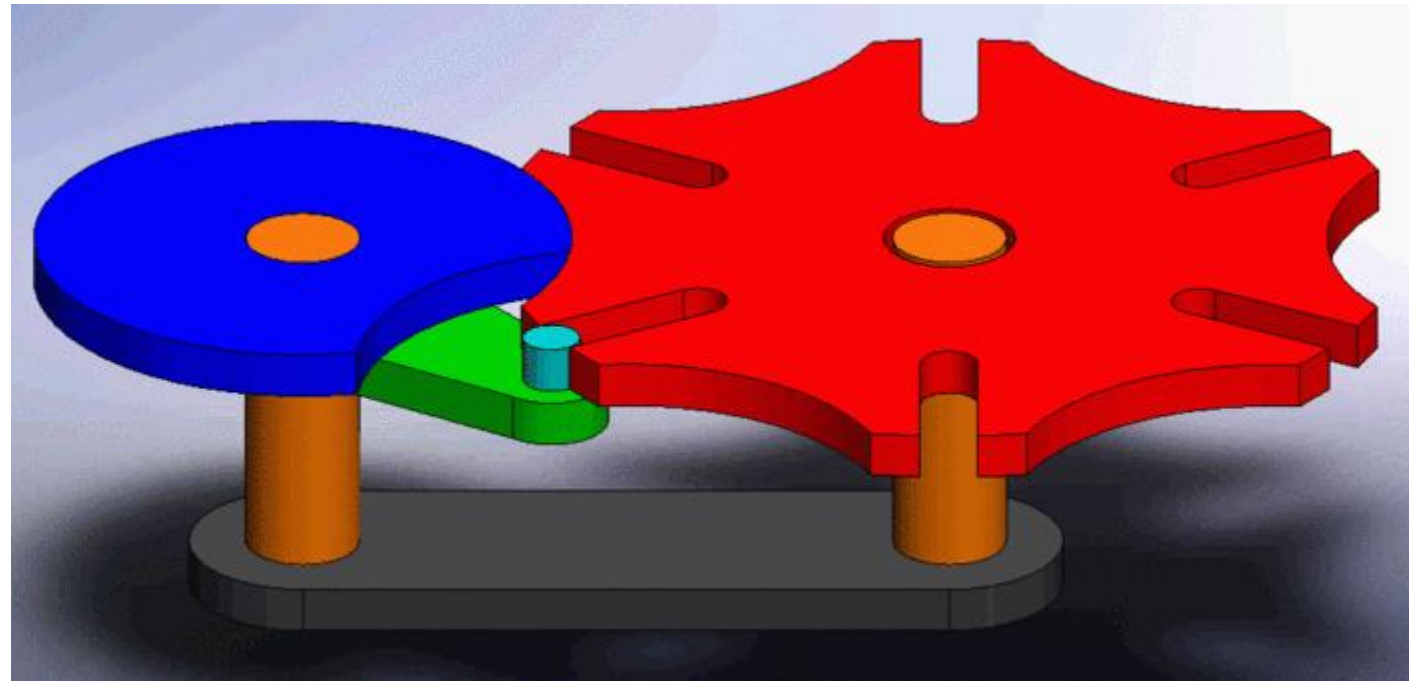
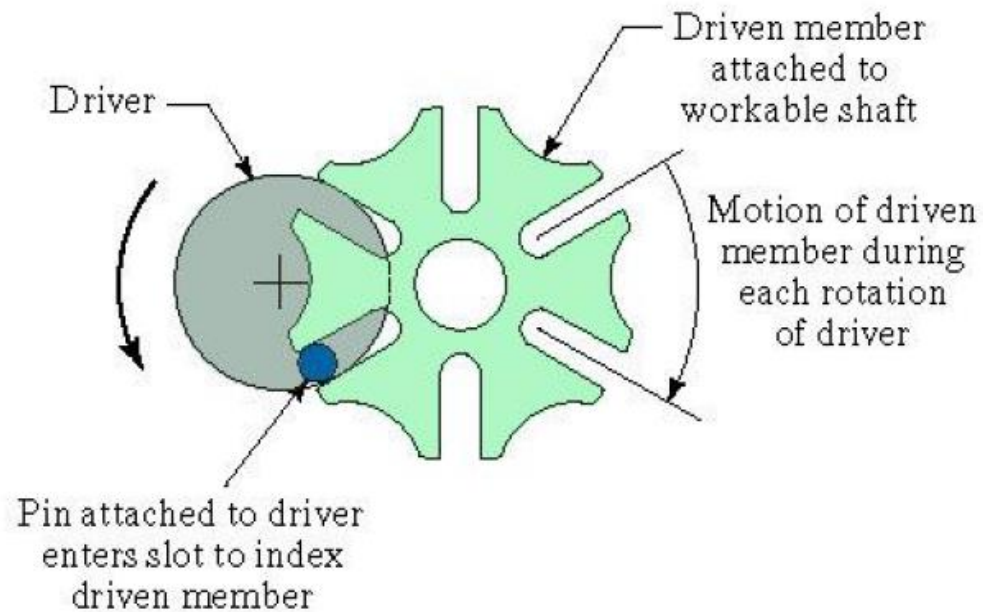
b. Ratchet and pawl

- A ratchet is a device that allows linear or rotary motion in only one direction, while preventing motion in the opposite direction.
- Ratchets consist of a gearwheel and a pivoting spring loaded finger called a pawl that engages the teeth



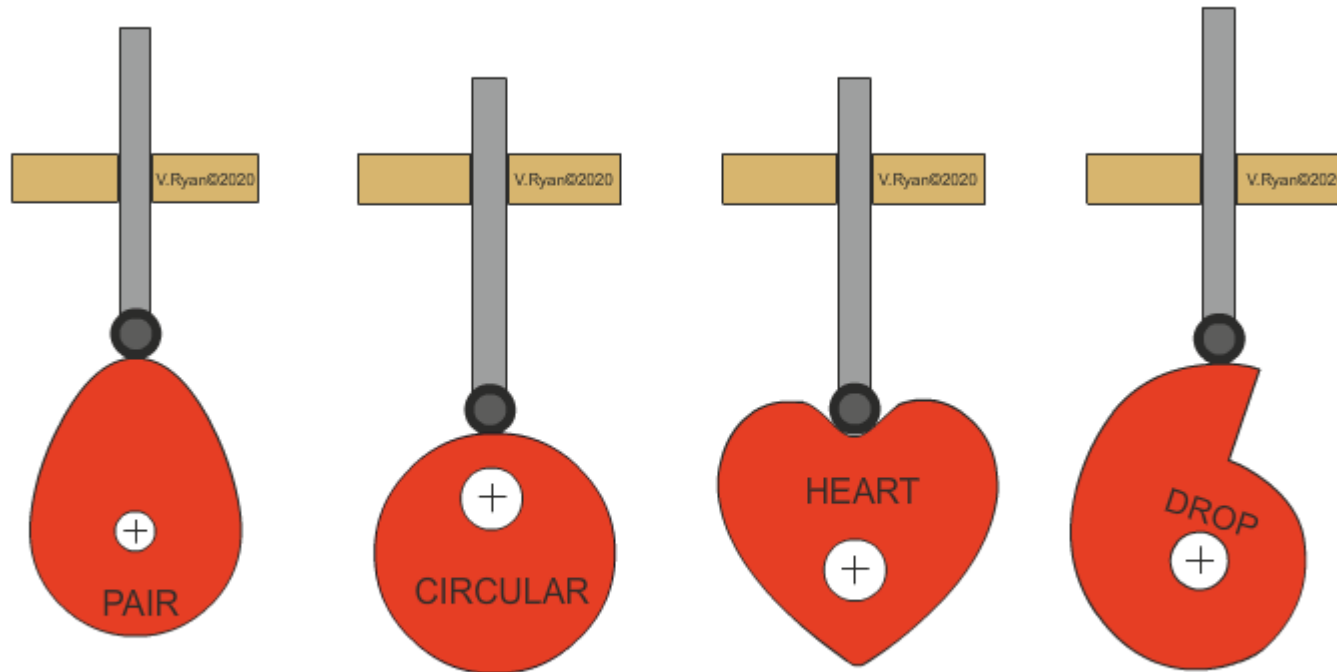
c. Geneva mechanism

- The two previous mechanisms convert a linear motion into a rotational motion.
- The Geneva mechanism uses a continuously rotating driver to index the table



d. CAM Mechanisms

- Various forms of cam mechanism provides accurate and reliable method of indexing the dial

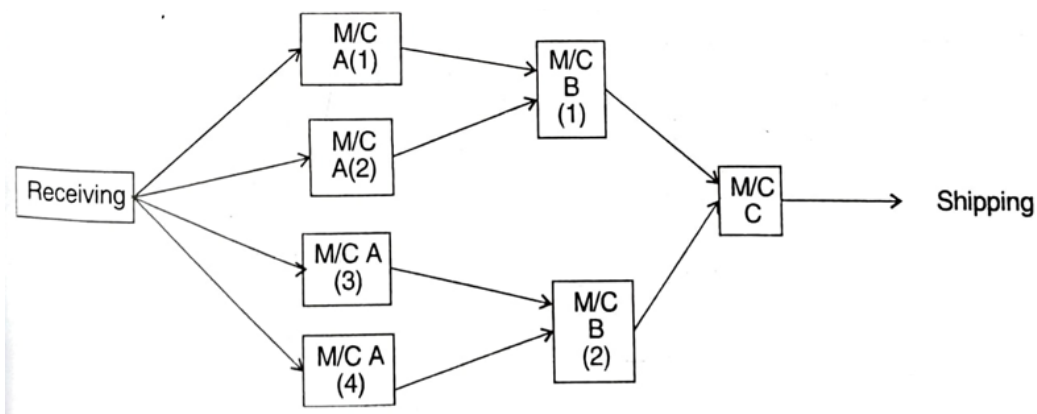


Line Balancing

Line balancing in a layout means balancing the production line or an assembly line. The main objective of line balancing is to distribute tasks evenly over the workstation so that idle time of man and machines is minimized.

Suppose, there are three machines (work stations) A, B and C which can process 25, 50 and 100 pieces per hour respectively and the pieces flow from A to B to C (precedence constraint). Since, A has minimum capacity, i.e. of processing 25 pieces per hour, work station or machine B will remain idle for 50% of its time and machine C will also remain idle for 50% of its time. Such a layout will be unbalanced one and the production time needs to be balanced.

The actual production in the line will be decided on the basis of the machine with the maximum production capacity. The production capacity of the other machines in the line will be adjusted through the increase in the number of machines. In this case, machine C produces 100 units per hour, hence 100 units will be manufactured per hour with the help of two machines B and from machine A. Thus, the line balancing for the production 100 units per hour will be done by arranging four machines of type A, two machines of type B with one machine of type C. Thus the idle capacity will be totally eliminated. The arrangement is as shown in figure:



Two important concepts in line balancing are the separation of the total work content into minimum rational work elements and the precedence constraints that must be satisfied by these elements.

Minimum Rational Work Elements - A minimum rational work element is a small amount of work that has a specific limited objective, such as adding a component to the base part, joining two components, or performing some other small portion of the total work content. A minimum rational work element cannot be subdivided any further without loss of practicality. For example, drilling a through-hole in a piece of sheet metal or fastening two machined components together with a bolt or screw would be defined as minimum rational work elements. It makes no sense to divide these tasks into smaller elements of work.

Precedence Constraints - In addition to the variation in element times that make it difficult to obtain equal service times for all stations, there are restrictions on the order in which the work elements can be performed. Some elements must be done before others. For example, to create a threaded hole, the hole must be drilled before it can be tapped.

Precedence constraints can be presented graphically in the form of a precedence diagram, which is a network diagram that indicates the sequence in which the work elements must be performed. Work elements are symbolized by nodes, and the precedence requirements are indicated by arrows connecting the nodes. The sequence proceeds from left to right.

$$T_{wc} = \sum T_e$$

$$P_p = \frac{D_a}{50 S_w H_{sh}}$$

$$T_c = \frac{60E}{P_p}$$

$$W^* = \text{minimum integer} \geq \frac{T_{wc}}{T_c}$$

$$T_s = \max(T_e) \leq T_c - T_r$$

$$E_b = \frac{T_{wc}}{W T_s}$$

$$d = \frac{W T_s - T_{wc}}{W T_s}$$

d - balance delay

*** A perfect line balance $E_b = 100\% \Rightarrow E_b + d = 1$

T_{wc} - total work content time

T_e - work element time

$\sum T_e$ - sum of work element time

D_a - Annual demand (units/yr)

P_p - avg. hourly production rate (units/hr)

50 weeks / year

OR
52 weeks / year

S_w - number of shifts / week

H_{sh} - Hr. / shifts

T_c - cycle time (min/cycle)

E - uptime efficiency for the line

P_p - production rate

W^* - Theoretical value of min. no. of workers.

T_r - Repositioning time

T_s - start time or service time

*** if T_r not given

$$\Rightarrow T_s = T_c$$

E_b - balance efficiency

W - Actual number of workers

(min. one worker per station)

A Problem for Line Balancing

A small electrical appliance is to be produced on a single-model assembly line. The work content of assembling the product has been reduced to the work elements listed in Table 15.4. The table also lists the times for each element and the precedence order in which they must be performed. The line is to be balanced for an annual demand of 100,000 units/yr. The line will operate 50 wk/yr, 5 shifts/wk, and 7.5 hr/shift. There will be one worker per station. Previous experience suggests that the uptime efficiency for the line will be 96%, and repositioning time lost per cycle will be 0.08 min. Determine (a) total work content time T_{wc} , (b) required hourly production rate R_p to satisfy the annual demand, (c) cycle time T_c , (d) theoretical minimum number of workers required on the line, and (e) service time T_s to which the line must be balanced.

No.	Work Element Description	T_{ek} (min)	Must be Preceded By
1	Place frame in work holder and clamp	0.2	–
2	Assemble plug, grommet to power cord	0.4	–
3	Assemble brackets to frame	0.7	1
4	Wire power cord to motor	0.1	1, 2
5	Wire power cord to switch	0.3	2
6	Assemble mechanism plate to bracket	0.11	3
7	Assemble blade to bracket	0.32	3
8	Assemble motor to brackets	0.6	3, 4
9	Align blade and attach to motor	0.27	6, 7, 8
10	Assemble switch to motor bracket	0.38	5, 8
11	Attach cover, inspect, and test	0.5	9, 10
12	Place in tote pan for packing	0.12	11

Solution: (a) The total work content time is the sum of the work element times in Table

$$T_{wc} = 4.0 \text{ min}$$

(b) Given the annual demand, the hourly production rate is

$$R_p = \frac{100,000}{50(5)(7.5)} = 53.33 \text{ units/hr}$$

(c) The corresponding cycle time T_c with an uptime efficiency of 96% is

$$T_c = \frac{60(0.96)}{53.33} = 1.08 \text{ min}$$

(d) The theoretical minimum number of workers

$$w^* = \text{Min Int } \geq \frac{4.0}{1.08} = 3.7 \text{ rounded up to 4 workers}$$

(e) The available service time against which the line must be balanced is

$$T_s = 1.08 - 0.08 = 1.00 \text{ min}$$

Line balancing algorithms

The objective in line balancing is to distribute the total workload on the assembly line as evenly as possible among the workers. several algorithms to solve the line balancing problem.

- (1) largest candidate rule
- (2) Kilbridge and Wester method
- (3) ranked positional weights method

Largest Candidate Rule

Step – 1

Draw the Precedence diagram

Step - 2

work elements are arranged in descending order according to their T_e (work element time) values.

Step -3

Assign elements to the first workstation by starting at the top of the list and selecting the first element that satisfies precedence requirements and does not cause the total sum of T_e at that station to exceed the allowable T_s (station time).

When an element is selected for assignment to the station, start back at the top of the list (descending order data in step - 1 and Precedence diagram) for subsequent assignments. when no more elements can be assigned without exceeding T_s , then proceed to the next station

Step-4

Repeat the steps 1 and 2 for as many additional stations as necessary until all elements have been assigned.

Problem

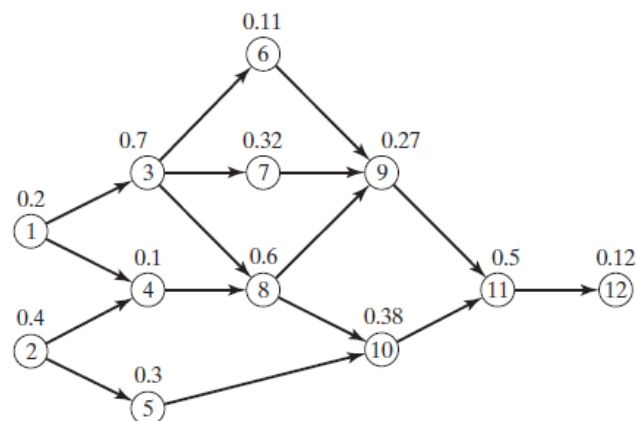
A new small electrical appliance is to be assembled on a production flow line. The total job of assembling the product has been divided into minimum rational work elements. The industrial engineering department has developed time standards based on previous similar jobs. This information is given in the table below. In the right-hand column are the immediate predecessors for each element as determined by precedence requirements. Assuming a cycle time of 1.0 min, find out the balance delay. Solve above the problems using Largest Candidate Rule

Element No.	Work element description	T_e (mins)	Must be preceded by
1	Place frame on work holder and clamp	0.2	---
2	Assemble plug, grommet to power cord	0.4	---
3	Assemble brackets to frame	0.7	1
4	Wire power cord to motor	0.1	1,2
5	Wire power cord to switch	0.3	2
6	Assemble mechanism plate to bracket	0.11	3
7	Assemble blade to bracket	0.32	3
8	Assemble motor to bracket	0.6	3,4
9	Align blade and attach to motor	0.27	6,7,8
10	Assemble switch to motor bracket	0.38	5,8
11	Attach cover, inspect, and test	0.5	9,10
12	Place in tote pan for packing	0.12	11

Solution

Step – 1

Precedence diagram for the above problem. Nodes represent work elements, and arrows indicate the sequence in which the elements must be done. Element times are shown above each node.



Step -2

Work Elements Arranged According to T_e Value for the Largest Candidate Rule

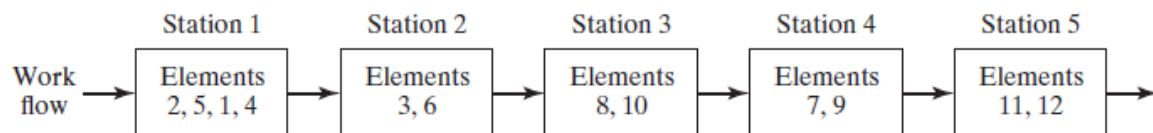
Work Element	T_{ek} (min)	Preceded By
3	0.7	1
8	0.6	3, 4
11	0.5	9, 10
2	0.4	—
10	0.38	5, 8
7	0.32	3
5	0.3	2
9	0.27	6, 7, 8
1	0.2	—
12	0.12	11
6	0.11	3
4	0.1	1, 2

Step – 3

Work Elements Assigned to Stations According to the Largest Candidate Rule

Station	Work Element	T_{ek} (min)	Station Time (min)
1	2	0.4	1.0
	5	0.3	
	1	0.2	
	4	0.1	
2	3	0.7	0.81
	6	0.11	
	8	0.6	
3	10	0.38	0.98
	7	0.32	
4	9	0.27	0.59
	11	0.5	
5	12	0.12	0.62

Based on the assignment the work flow is as given below



$$\text{Balance delay } d = \frac{wT_c - T_{wc}}{wT_c} = \frac{(5 * 1) - 4}{5 * 1} = 0.2 = 20\%$$